

Online Sales Data Analysis

Capstone Project Report

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Tool Used: Microsoft Power BI | Dataset: Online Retail Sales

1. Project Overview

This capstone project focuses on performing an end-to-end data analysis on an Online Sales dataset using Microsoft Power BI. The primary objective was to explore, clean, and visualise sales data to derive meaningful business insights that can assist decision-makers in identifying revenue trends, top-performing product categories, regional performance, and customer payment preferences.

The dataset comprised transactional records including fields such as Order Date, Product Name, Product Category, Units Sold, Unit Price, Total Revenue, Region, and Payment Method. The analysis covered data spanning multiple months and regions, enabling a comprehensive view of the business performance.

2. Methodology and Steps Followed

- **Step 1 – Data Collection & Import:**

The Online Sales dataset (CSV format) was imported into Power BI Desktop using the 'Get Data' option. The dataset contained fields such as order date, product details, revenue, region, and payment method.

- **Step 2 – Data Cleaning (Power Query Editor):**

The dataset was cleaned using Power BI's built-in Power Query Editor. Tasks included removing duplicate records, handling null/missing values, correcting data types (e.g., converting date columns to Date format and revenue columns to Decimal Number), and renaming columns for clarity.

- **Step 3 – Data Transformation:**

Calculated columns and measures were created using DAX (Data Analysis Expressions). Key measures included Total Revenue (SUM), Total Units Sold (SUM), and Average Order Value (DIVIDE). A Month column was extracted from the Order Date for time-series analysis.

- **Step 4 – Data Modelling:**

Relationships between tables (if multiple) were defined in the Model View to ensure accurate cross-filtering across visuals. A star schema approach was followed where applicable.

- **Step 5 – Dashboard Design & Visualisation:**

An interactive Power BI dashboard was designed with the following visuals: KPI Cards (Total Revenue, Total Units Sold, Average Order Value), a Bar Chart for Revenue by Product Category, a Line Chart for Monthly Revenue Trend, a Donut Chart for Revenue by Payment Method, a Pie Chart for Revenue by Region, and a Horizontal Bar Chart for Top Products by Revenue.

- **Step 6 – Formatting & Publishing:**

The dashboard was formatted with a consistent colour theme. Titles, labels, and legends were added for clarity. Slicers/filters were considered to allow dynamic exploration of data.

3. Key Insights from the Dashboard

The following insights were derived from the Power BI dashboard visualisations:

Metric	Value / Observation
Total Revenue	\$51.17K across all product categories and regions
Total Units Sold	268 units sold across all orders
Average Order Value	\$426.42 per order on average
Top Product Category	Electronics generated the highest revenue (~\$35K)
Monthly Revenue Trend	Revenue peaked in January (\$9.9K) and declined to \$3.8K by July
Top Region	North America led with 45.73% of total revenue (\$36.84K)
Preferred Payment Method	Credit Card was most used (63.51% — \$51.17K contribution)
Best-Selling Product	Canon EOS R5 Camera ranked #1 in revenue by product name

- Electronics and Home Appliances together accounted for the majority of total revenue, highlighting the dominance of high-value tech products in the sales mix.
 - A consistent downward trend in monthly revenue from January to July suggests possible seasonality or inventory/marketing issues in the latter half of the period.
 - North America is the primary revenue-driving region, with Europe and Asia contributing approximately 17.78% and a minimal share respectively.
 - Credit Card was the overwhelmingly preferred payment method, suggesting the platform's customer base favours card-based transactions over digital wallets.
 - Premium products such as the Canon EOS R5 Camera, LG OLED TV, and MacBook Pro 16-inch were the top revenue contributors at the individual product level.
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4. Power BI Dashboard

The following screenshot illustrates the final interactive dashboard created in Microsoft Power BI, consolidating all key metrics and visual representations of the Online Sales dataset:

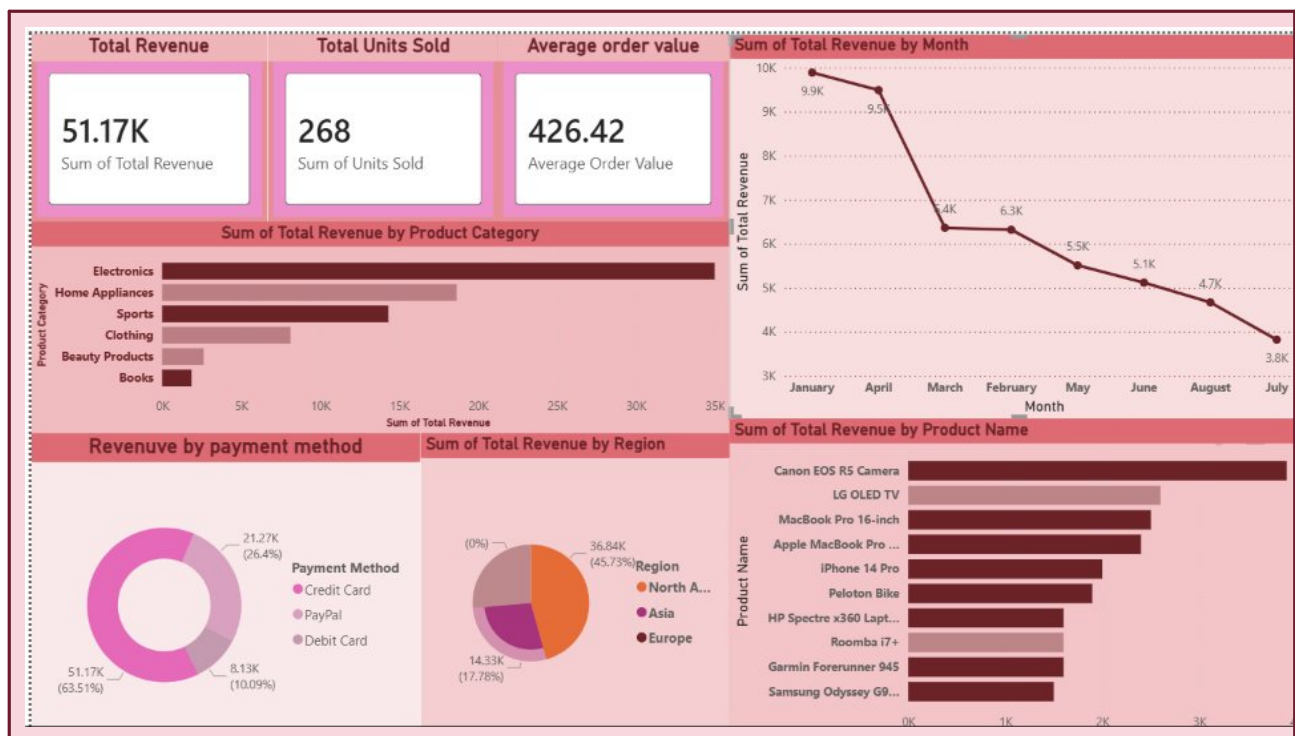


Figure 1: Power BI Dashboard — Online Sales Data Analysis (by Rithikha)

5. Conclusion

This project successfully demonstrated the complete data analysis workflow — from raw data ingestion and cleaning to transformation, modelling, and interactive visualisation using Microsoft Power BI. The dashboard provides a single-glance view of business performance across revenue, products, regions, and payment methods.

The project strengthened practical skills in Power Query, DAX measure creation, and dashboard design principles. The insights generated can directly support strategic decisions such as targeted marketing in underperforming months, regional expansion planning, and inventory management for high-demand product categories.